# PM Digest; Multi-Agent Orchestration

***Slack + Gmail + Calendar → one synthesized daily digest, built as an n8n multi-agent workflow.***

## Problem

A PM's actual workload; decisions, deadlines, blockers, follow-ups; is scattered across Slack, Gmail, and Calendar. Checking each separately is slow, and cross-source connections (a deadline mentioned in email that's also a Slack thread topic) are easy to miss.

## Approach & Key Decisions

- Three-layer design: Control (Orchestrator/Guardrails/State) → Execution (3 parallel source agents) → Reliability (error handling).
- Aggregate nodes batch multi-item sources (Slack messages, emails) into a single LLM call per source, instead of running the agent once per item; fixes fragmented output and duplicate downstream execution.
- An early idempotency guard compares today's date to the last logged digest date, skipping all agent/LLM calls entirely on a same-day re-run rather than discarding duplicate output at the end.
- Source agents run on gpt-4o-mini (fast, cheap, structured extraction); the Synthesizer is wired for gpt-4o since deduping and prioritizing across 3 sources is a harder reasoning task; model choice is swappable per node.

## What It Does

- Slack Agent: searches a channel, extracts decisions/deadlines/blockers/actions, flags VIP mentions.
- Gmail Agent: filters and triages emails for decisions, approvals, deadlines, risks, stakeholder asks.
- Calendar Agent: summarizes today's events and prep needed.
- Synthesizer Agent: merges all 3, deduplicates, tags [New]/[Recurring]/[VIP], sorts by urgency.
- Output: posts the digest to Slack and logs it to Google Sheets, feeding the next run's history.

## Results & Learnings

- The three-layer split made debugging tractable; failures were always isolable to config, a single agent, or the merge/synthesis step.
- The dominant bug class was item-count mismatches across parallel branches, not AI reasoning quality; n8n runs a node once per item by default, and unbatched sources silently multiplied every downstream step.
- Next: the Calendar branch still runs its agent once per event rather than being aggregated; fine at low volume, same fan-out risk exists on a busier day.

## Tech Stack

| Component | Choice |
|---|---|
| Orchestration platform | n8n (self-hosted / n8n.cloud) |
| LLM; source agents | OpenAI gpt-4o-mini |
| LLM; Synthesizer | OpenAI (wired for gpt-4o; swappable) |
| Agent framework | @n8n/n8n-nodes |
| Sources | Slack API, Gmail API, Google Calendar API |
| State / history store | Google Sheets |
| Output | Slack channel post |